P1-C Group 3: analytical multilayer script, validation report

EM wave penetration in layered tissue (2.45 GHz team). Prepared by Yusuf Uzun, 2026-10-26. Evidence for Checkpoint 1 (script part). Scope: the Python script only. No Feko result is included yet.

1. Checks against the pass criteria of the workbook

Check	Reference	Pass criterion	Result	Pass?
Script: single interface	Fresnel formula, air/muscle, 4 frequencies	4 significant digits	equal at all 4 frequencies; max |Γ| difference 2.2e-16	PASS
Script: λ/4 matching layer	Textbook, lossless layer, η1 = √(η0·η2)	|Γ| < 1e-6	|Γ| = 2.2e-16	PASS
Penetration depth in muscle (script vs Week 1 table)	Week 1 slide: 2.23 cm at 2.45 GHz	within 2 %	δ = 22.33 mm (0.13 % from 22.3 mm)	PASS
Script: power conservation	1 − |Γ|² = Σ P_abs/S_inc + P_trans/S_inc	small residual	max error 7.2e-10	PASS
Tissue media vs Week 1 table	Week 1 slide 17 (ε′, σ at 4 frequencies)	within 1 %	worst 0.90 % (fat eps_r at 5800 MHz)	PASS
Tissue media vs course media pack	Course files Muscle / Skin_dry / Fat_not_infiltrated (216 points each, 1 MHz to 20 GHz)	within 1 %	worst 0.0031 % at the file's own frequencies	PASS
Feko vs script	|Γ|, per-layer absorption	within 3 %	not done yet (no Feko result)	open
Power balance (dipole case)	P_acc = P_rad + P_abs	error < 5 %	not done yet (Week 6)	open

2. Test 1: air/muscle interface vs Fresnel

Muscle half-space, Gabriel properties at each frequency, normal incidence. Reference Γ = (1 − n)/(1 + n) with n = √ε*. Compared as complex numbers (difference below 1e-9).

f (MHz)	script |Γ|	Fresnel |Γ|	1 − |Γ|² (power entering)	4 digits
403	0.790354	0.790354	37.53 %	PASS
915	0.770809	0.770809	40.59 %	PASS
2450	0.762460	0.762460	41.87 %	PASS
5800	0.756698	0.756698	42.74 %	PASS

3. Test 2: lossless quarter-wave layer

Layer ε1 = 3 (λ1/4 thick) on a lossless ε2 = 9 half-space, η1 = √(η0·η2). At the design frequency \|Γ\| = 2.2e-16. As a counter-check, at twice the frequency the layer is λ/2 thick and the stack must behave like the bare interface: \|Γ\| = 0.500000 versus 0.500000 for the bare interface.

4. Power conservation, skin 1.5 mm / fat 8 mm / muscle half-space

Absorbed power per layer is the drop of the Poynting flux across the layer. It is cross-checked by integrating ½σ|E|² over the layer (last column).

f (MHz)	|Γ|	1 − |Γ|²	Σ P_abs/S_inc	P_trans/S_inc	conservation error	Poynting vs ∫½σ|E|²
403	0.7665	0.41250	0.41250	0.0e+00	1.4e-10	3.3e-07
915	0.6515	0.57561	0.57561	0.0e+00	2.5e-10	3.3e-07
2450	0.5231	0.72636	0.72636	0.0e+00	6.3e-10	3.3e-07
5800	0.8669	0.24851	0.24851	0.0e+00	7.2e-10	3.3e-07

5. Penetration depth and the finite-slab bias

Muscle half-space at 2.45 GHz: analytic δ = 22.330 mm, fit of ln|E| (Week 1 procedure, 5 mm skipped at each face) = 22.330 mm (+0.00 %). For a 60 mm muscle slab backed by air, as in a Feko model, the back-face reflection (|Γ| about 0.76 for muscle/air) adds a ripple: the same fit on the exact field gives 22.80 mm (+2.12 %), which is close to the 2 % limit. Skipping 1.5 δ at the bottom of the slab gives 22.43 mm (+0.47 %). A Feko δ check should therefore use both windows.

6. Unit tests (pytest)

Result: 9 passed in 0.59s. Environment: Python 3.13.16, numpy 2.5.3, matplotlib 3.11.2.

Test	Result
test_fresnel_4_digits	PASSED
test_fresnel_matches_week1_slide	PASSED
test_quarter_wave_match	PASSED
test_power_conservation	PASSED
test_field_continuity	PASSED
test_gabriel_vs_week1_table	PASSED
test_penetration_depth_fit	PASSED
test_penetration_depth_fit_finite_slab	PASSED
test_dipole_reference	PASSED

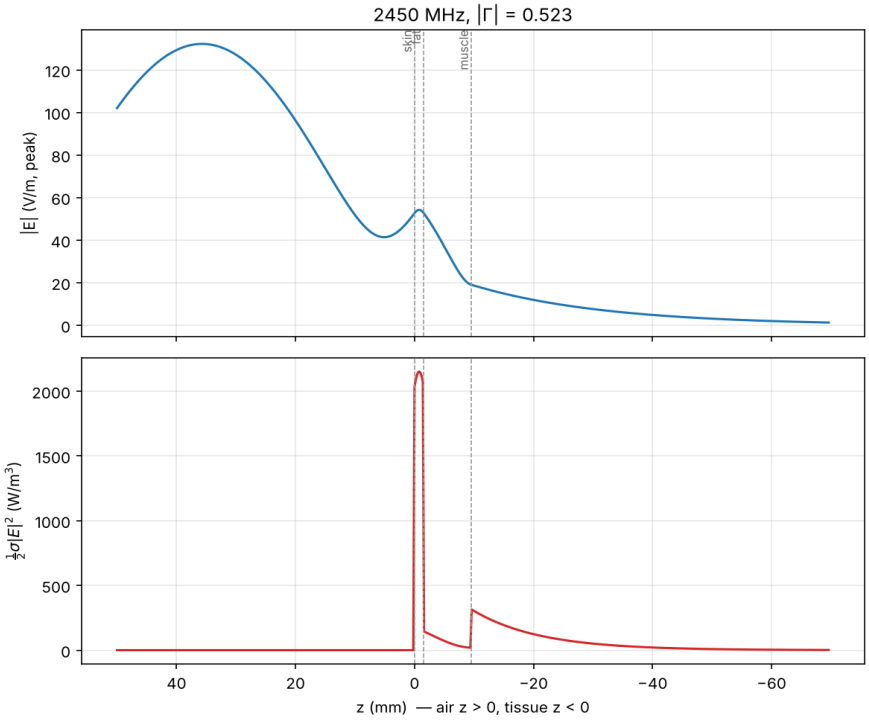


Figure 1. |E|(z) and ½σ|E|²(z) at 2.45 GHz for skin 1.5 / fat 8 / muscle half-space (10 W/m² incident; air on the left).

7. Limitations and open items

• All results come from the script; the Feko models (plane-wave PBC cell, mesh convergence) and the model log are not done yet. Checkpoint 1 also needs those.
• The tissue properties are computed from the published Gabriel (1996) four-term Cole-Cole parameters. They agree with the course media pack files to 0.0031 % at the file frequencies, and with the Week 1 table to 0.90 %.
• The tests check the script against analytical references and the lecture numbers, so they cannot reveal a convention mismatch with Feko; that is what the Feko vs script comparison is for.